Roth & Sant'Anna (2023): A Monte Carlo Study

Roberto Diaz Hernandez Riana Ramonjamanana Charlene Ramos

March 18, 2026

I. Summary

Difference-in-differences (DiD) is one of the most widely used research designs in applied economics, but its validity rests on the parallel trends (PT) assumption: that treated and comparison groups would have followed the same trajectory absent treatment. A recurring practical problem is that PT is not invariant to the scale on which outcomes are measured. A researcher who finds no pre-treatment divergence in levels may find one in logs, or vice versa. Roth and Sant'Anna (2023) ask when this ambiguity can be resolved, characterizing exactly when PT holds under all strictly monotonic transformations of the outcome.

This paper connects two central themes of the class. First, it extends the broader critique of PT fragility, alongside the TWFE problems identified by Goodman-Bacon (2021) and de Chaisemartin and D'Haultfœuille (2020), by showing that even a correctly specified DiD can mislead when PT does not carry over across scales. Second, it motivates the changes-in-changes (CiC) model of Athey and Imbens (2006) as a natural robust alternative: since functional form robustness requires PT at the CDF level, the CiC model sidesteps the functional form problem by identifying the full counterfactual distribution under structural assumptions. The paper thus both diagnoses a limitation of mean-based DiD and points toward the distributional methods that address it.

Contribution

Proposition 3.1 establishes that PT is robust to all monotonic transformations if and only if parallel trends holds for the full CDF of untreated potential outcomes, not merely their means. Proposition 3.2 characterizes when this holds: untreated potential outcomes must admit a mixture representation combining a common time-varying component and a group-specific time-invariant component. Outside this structure, functional form invariance will generally fail, and researchers must justify their choice of scale directly.

Relevance to Applied Research

The absence of pre-trends in a chosen functional form does not validate PT under alternative transformations. However, the distributional PT condition generates a falsification test: the implied counterfactual CDF for the treated group is fully identified from observables, and if it fails monotonicity, functional form invariance is rejected. This diagnostic is easy to implement before committing to a particular outcome transformation.

II. Theoretical properties of functional form invariance in difference-in-differences

The main theoretical contribution of Roth and Sant'Anna (2023) is to characterize when the standard difference-in-differences parallel trends assumption is invariant to monotonic transformations of the outcome variable. In the two-period difference-in-differences setting, identification of the average treatment effect on the treated relies on equality of changes of the average untreated outcome across treated and comparison groups. Formally, the standard parallel trends assumption requires

$$\mathbb{E}[Y_{i1}(0) \mid D_i = 1] - \mathbb{E}[Y_{i0}(0) \mid D_i = 1] = \mathbb{E}[Y_{i1}(0) \mid D_i = 0] - \mathbb{E}[Y_{i0}(0) \mid D_i = 0].$$

Under this condition, the average treatment effect on the treated is identified by the standard difference-in-differences estimator. The paper studies when this identification result remains valid after replacing the outcome by an arbitrary strictly monotonic transformation $g(Y)$. This stronger requirement implies

$$\mathbb{E}[g(Y_{i1}(0)) \mid D_i = 1] - \mathbb{E}[g(Y_{i0}(0)) \mid D_i = 1] = \mathbb{E}[g(Y_{i1}(0)) \mid D_i = 0] - \mathbb{E}[g(Y_{i0}(0)) \mid D_i = 0]$$

for every strictly monotonic function g for which the expectations are finite.

The first main theoretical result, proposition 3.1, shows that this invariance property holds if and only if a stronger distributional restriction is satisfied. Specifically, parallel trends must hold for the full cumulative distribution function of untreated potential outcomes:

$$F_{Y_{i1}(0)|D_i=1}(y) - F_{Y_{i0}(0)|D_i=1}(y) = F_{Y_{i1}(0)|D_i=0}(y) - F_{Y_{i0}(0)|D_i=0}(y)$$

for every $y \in \mathbb{R}$.

If the difference between cumulative distribution functions is identical across groups, then equality holds for every strictly monotonic transformation g. Conversely, if invariance holds for all monotonic transformations, then choosing particular functions g implies equality of the distributional differences pointwise.

An important implication is that when the outcome variable is continuous, parallel trends for cumulative distribution functions is equivalent to parallel trends for probability density functions almost everywhere. Denoting densities by $f_{Y_{it}(0)|D_i=d}(y)$, the condition becomes

$$f_{Y_{i1}(0)|D_i=1}(y) - f_{Y_{i0}(0)|D_i=1}(y) = f_{Y_{i1}(0)|D_i=0}(y) - f_{Y_{i0}(0)|D_i=0}(y)$$

for almost every y. In this case, functional form invariance requires that the full untreated density shifts over time identically across groups.

The second theoretical result, Proposition 3.2, characterizes the distributions that satisfy this condition. Assuming that untreated outcome distributions admit Radon–Nikodym densities with respect to a common dominating measure, the paper shows that the untreated potential outcomes can be represented as

$$F_{Y_{it}(0)|D_i=d}(y) = \theta G_t(y) + (1-\theta) H_d(y),$$

where $G_t(y)$ depends only on time, $H_d(y)$ depends only on treatment status, and $\theta \in [0,1]$.

The parameter θ determines the economic interpretation of the decomposition. When $\theta = 1$, the untreated distributions coincide across groups in every period:

$$F_{Y_{it}(0)|D_i=1}(y) = F_{Y_{it}(0)|D_i=0}(y),$$

which corresponds to a setting of effective random assignment. In this case, treatment and comparison groups share the same untreated distribution before and after treatment.

When $\theta = 0$, untreated distributions are group-specific but invariant over time:

$$F_{Y_{i1}(0)|D_i=d}(y) = F_{Y_{i0}(0)|D_i=d}(y),$$

for $d \in \{0,1\}$. Identification then follows because untreated outcomes are stationary within each group.

The intermediate case $0 < \theta < 1$ corresponds to a mixture structure in which each group contains two components. A fraction θ follows a common time-varying distribution $G_t(y)$, while the remaining fraction $1-\theta$ follows a group-specific distribution $H_d(y)$ that does not change over time. Formally,

$$F_{Y_{it}(0)|D_i=d}(y) = \theta G_t(y) + (1-\theta) H_d(y)$$

implies that both common time variation and permanent group heterogeneity coexist.

A key implication of Proposition 3.2 is that invariance to functional form requires highly

restrictive distributional structure whenever treatment is not randomly assigned. In many empirical applications, these conditions are unlikely to hold exactly, which implies that researchers must justify the choice of functional form directly rather than assuming robustness across transformations.

The theoretical results also imply a directly testable restriction. Rearranging the cumulative distribution condition yields the implied untreated counterfactual distribution for the treated group in the post-treatment period:

$$F_{Y_{i1}(0)|D_i=1}(y) = F_{Y_{i0}(0)|D_i=1}(y) + F_{Y_{i1}(0)|D_i=0}(y) - F_{Y_{i0}(0)|D_i=0}(y).$$

This expression is the distributional analogue of the standard difference-in-differences estimator. The right-hand side is fully identified from observable distributions: the pre-treatment distribution of the treated group, together with the pre-treatment and post-treatment distributions of the comparison group. Under the null hypothesis of functional form invariance, these observed components imply the untreated post-treatment distribution for treated units.

Because the left-hand side must be a valid cumulative distribution function, the right-hand side must satisfy the defining properties of a cumulative distribution function. In particular, it must remain weakly increasing in y. If the implied counterfactual distribution decreases over some range of the support, then the resulting function cannot represent a valid probability distribution. In that case, the distributional parallel trends condition is violated, and the null hypothesis that parallel trends is invariant to functional form is rejected.

This provides a practical falsification test: if the estimated counterfactual distribution implied by the data fails monotonicity, then the stronger distributional assumptions required for invariance across transformations do not hold.

III. Monte Carlo simulations

We evaluate the theoretical results of Roth and Sant'Anna (2023) through two Monte Carlo experiments. Both experiments are set in the two-period, two-group framework with zero treatment effect ($\tau_{\text{ATT}} = 0$). Any nonzero DiD estimate is attributable to the failure of parallel trends under the chosen functional form. All outcomes are drawn from log-normal distributions, which provide a natural testbed given their means are nonlinear functions of the distributional parameters. We set $n = 1{,}000$ observations per group-period cell and conduct $B = 2{,}000$ replications in each experiment. Inference is based on a Wald test at the 5% level, where the standard error is computed from the sample variances in each cell.

DGP 1: Favorable case — Invariant parallel trends

The first data-generating process satisfies the mixture representation of Proposition 3.2, ensuring that parallel trends is invariant to functional form. We generate counterfactual outcomes for each group-period cell as a 50–50 mixture of two log-normal components:

$$F_{Y_{it}(0)|D_i=d}(y) = 0.5 \cdot G_t(y) + 0.5 \cdot H_d(y),$$

where $G_t \sim \text{Lognormal}(2+t,\ 1)$ captures common time variation and $H_d \sim \text{Lognormal}(3+d,\ 1)$ captures permanent group heterogeneity. This gives us:

$$\begin{aligned}
G_0(y) &= \Phi\left(\frac{\ln y - 2}{1}\right) && \text{(pre-period time effect)} \\
G_1(y) &= \Phi\left(\frac{\ln y - 3}{1}\right) && \text{(post-period time effect)} \\
H_0(y) &= \Phi\left(\frac{\ln y - 3}{1}\right) && \text{(control group effect)} \\
H_1(y) &= \Phi\left(\frac{\ln y - 4}{1}\right) && \text{(treated group effect)}
\end{aligned}$$

with Φ as the standard normal CDF. Note that $\theta = 0.5$ throughout, so neither the time effect nor the group effect dominates.

Since the time-varying component G_t shifts identically across groups while the group-specific component H_d remains fixed over time, the distributional parallel trends condition of Proposition 3.1 is satisfied by construction. The expected values confirm that the DiD estimand equals zero in both levels and logs. In levels, $\mathbb{E}[Y|d,t] = 0.5\exp(\mu_1 + 0.5) + 0.5\exp(\mu_2 + 0.5)$, giving group means of 22.65 (control, pre), 33.12 (control, post), 51.10 (treated, pre), and 61.57 (treated, post); the DiD equals $(61.57-51.10)-(33.12-22.65) = 0$. In logs, $\mathbb{E}[\log Y|d,t] = 0.5\mu_1 + 0.5\mu_2$, giving 2.50, 3.00, 3.00, and 3.50 respectively, with a DiD of $(3.50 - 3.00) - (3.00 - 2.50) = 0$.

Table 1 reports the Monte Carlo results. In both levels and logs, the mean DiD estimate is centered near zero, the empirical bias is negligible relative to the standard deviation, and the rejection rate is close to the 5% level. This confirms that when the mixture representation is satisfied, the standard DiD estimator recovers an unbiased estimate of the ATT regardless of the functional form applied to the outcome.

The contrast between the two DGPs is visible at the distributional level. Figure 1 plots the population densities for all four group-period cells under each DGP. In the left panel (DGP 1), both groups exhibit the bimodal shape generated by the 50–50 mixture; the treated densities are shifted rightward relative to the control densities, but the overall shape

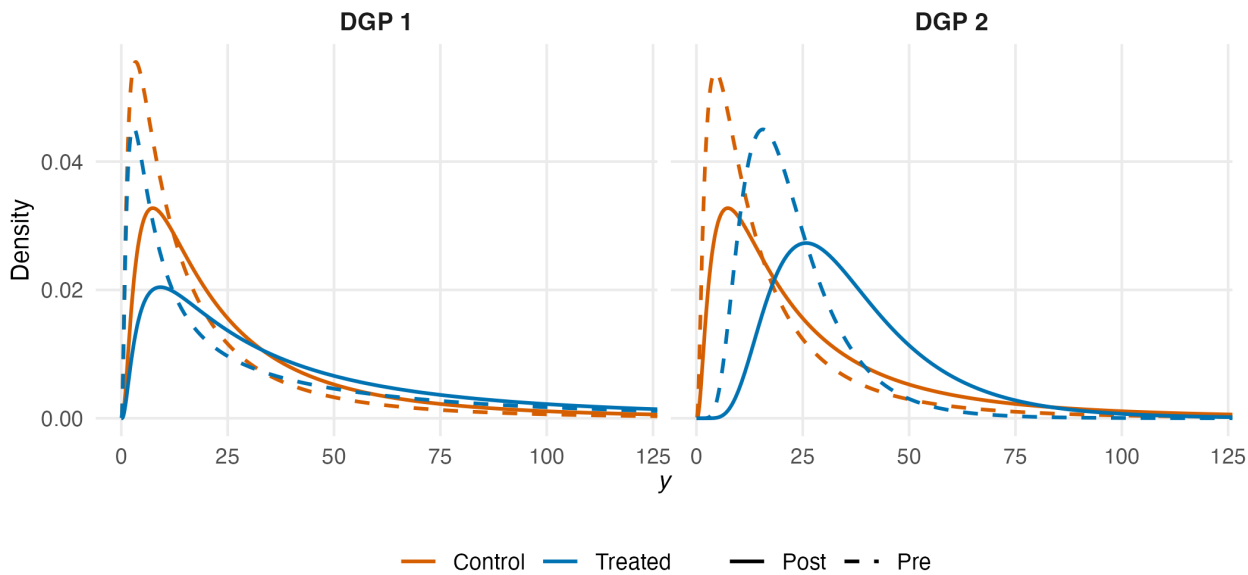
Table 1. Monte Carlo results: DGP 1 (favorable case)

$g(\cdot)$	True DiD	Mean $\hat{\tau}$	Bias	SD	Mean SE	Rej. Rate
Levels	0.000	0.051	0.051	4.258	4.487	0.031
Logs	0.000	-0.002	-0.002	0.071	0.074	0.036

Note. $B = 2{,}000$ replications, $n = 1{,}000$ per cell. True ATT $= 0$ under all transformations.

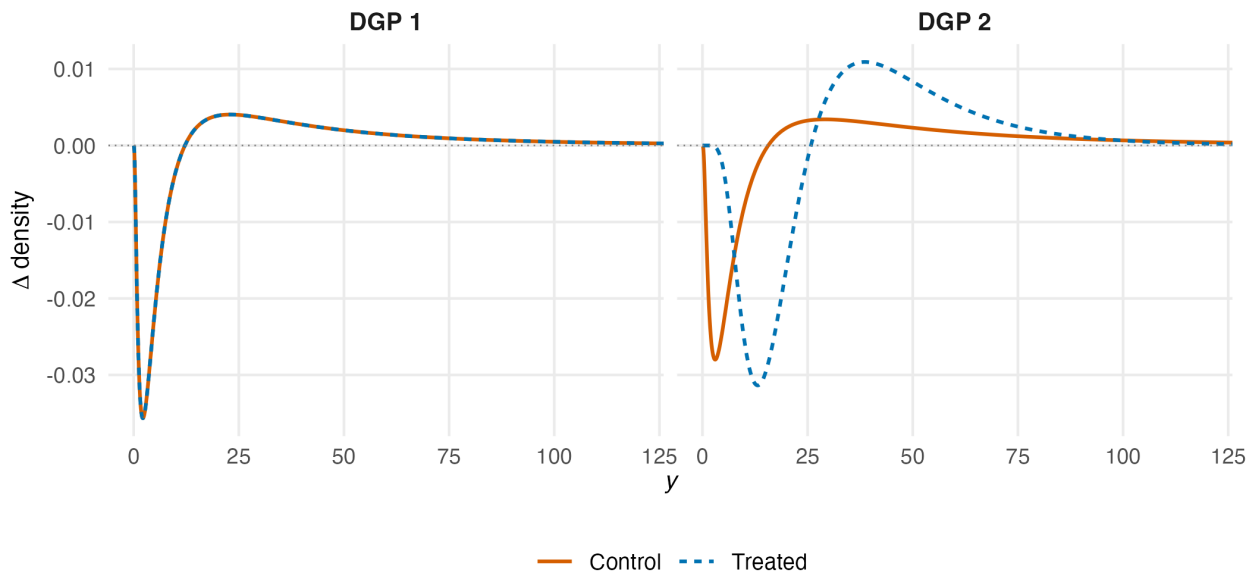
of each density is governed by the same pair of log-normal components. In the right panel (DGP 2), the control group's heavy-tailed density contrasts sharply with the treated group's concentrated density, foreshadowing the divergence in level-scale moments.

Figure 1. Population densities of untreated potential outcomes



A further diagnostic confirms that the change in density across periods is identical for the two groups. Because G_t shifts from $\ln(2,1)$ to $\ln(3,1)$ and affects both groups equally, while H_d contributes a fixed group-specific density component, the change-in-density function $f_{Y_{i1}(0)|D_i=d}(y) - f_{Y_{i0}(0)|D_i=d}(y)$ is the same for $d = 0$ and $d = 1$. This equality of density shifts is the pointwise analogue of distributional parallel trends and visually corroborates the invariance result.

Figure 2. Change in density across periodsfor control and treated groups



Note. Left: DGP 1 (curves overlap, confirming distributional parallel trends). Right: DGP 2 (curves diverge, signaling a violation).

DGP 2: Unfavorable case — Non-invariant parallel trends

The second DGP violates the mixture representation by allowing the treated and control groups to feature different scale parameters. Specifically, each group-period cell follows a log-normal distribution:

$$\begin{aligned}
&\text{Control} && \text{Pre-period: } Y_{i0}(0) \sim \text{Lognormal}(\mu = 2.5, \sigma = 1.0) \\
& && \text{Post-period: } Y_{i1}(0) \sim \text{Lognormal}(\mu = 3.0, \sigma = 1.0) \\
&\text{Treated} && \text{Pre-period: } Y_{i0}(0) \sim \text{Lognormal}(\mu = 3.0, \sigma = 0.5) \\
& && \text{Post-period (counterfactual): } Y_{i1}(0) \sim \text{Lognormal}(\mu = 3.5, \sigma = 0.5)
\end{aligned}$$

The key feature is that the control group has $\sigma = 1.0$ while the treated group has $\sigma = 0.5$. Although the log-mean shift is identical across groups ($\Delta\mu = 0.5$ for both), the nonlinear mapping from log-means to level-means implies that the level-scale changes differ. For log-normal variables, $\mathbb{E}[Y] = \exp(\mu + \sigma^2/2)$, so the level-scale group means are 20.09, 33.12, 22.76, and 37.52 for (control pre, control post, treated pre, treated post), respectively. In logs, the DiD estimand is $(3.5 - 3.0) - (3.0 - 2.5) = 0$, whereas in levels the DiD is $(37.52 - 22.76) - (33.12 - 20.09) = 1.74 \neq 0$.

Table 2 shows that the Monte Carlo results reflect this asymmetry precisely. In logs, the DiD estimator is unbiased: the mean estimate is -0.001, the bias is negligible, and the rejection rate of 0.051 is close to the 5% level. In levels, however, the estimator exhibits a systematic positive bias of 1.719, reflecting the mismatch between log-scale parallel trends

and level-scale moments. The rejection rate of 0.188 is nearly four times the nominal level, and as shown in Figure 5, this rate increases monotonically with sample size as the fixed bias dominates the shrinking standard error. A researcher who estimates a DiD model in levels would incorrectly conclude that the treatment had a large positive effect of approximately 1.72 units, when the true $\tau_{\text{ATT}} = 0$ under all transformations.

Table 2. Monte Carlo results: DGP 2 (unfavorable case)

$g(\cdot)$	True DiD	Mean $\hat{\tau}$	Bias	SD	Mean SE	Rej. Rate
Levels	0.000	1.719	1.719	1.795	1.758	0.188
Logs	0.000	-0.001	-0.001	0.051	0.050	0.051

Note. $B = 2{,}000$ replications, $n = 1{,}000$ per cell. True ATT $= 0$ under all transformations.

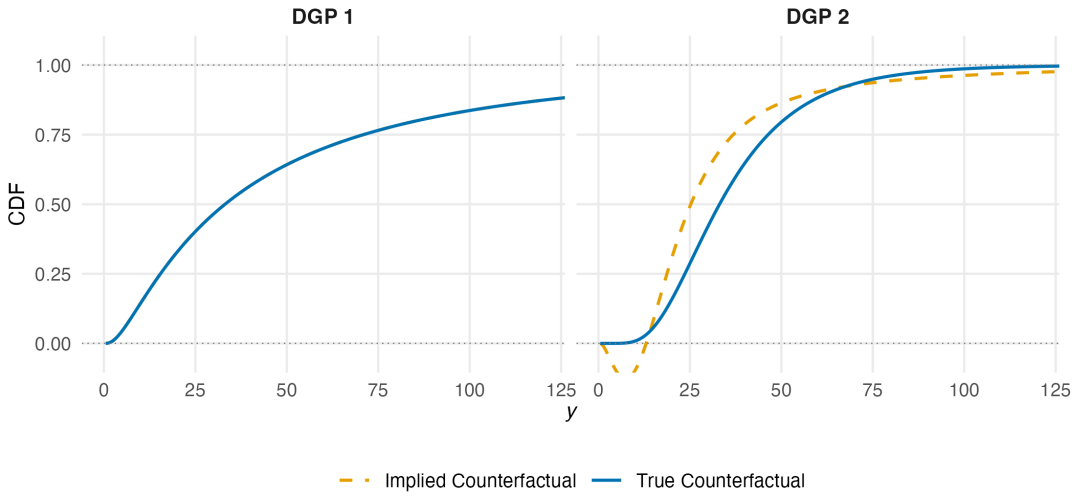
The source of the bias is visible in the density-change diagnostic. In DGP 1, the change in density between periods is identical for the control and treated groups, reflecting the common time-varying component G_t in the mixture. In DGP 2, the shapes of the density changes are markedly different. The control group's broad-tailed log-normal generates a more dispersed shift, whereas the treated group's concentrated density shifts more sharply. This divergence signals that the distributional parallel trends condition is violated.

We also apply the falsification test proposed in the paper. Under distributional parallel trends, the implied counterfactual CDF for treated units in the post-treatment period is

$$\hat{F}_{Y_{i1}(0)|D_i=1}(y) = F_{Y_{i0}(0)|D_i=1}(y) + F_{Y_{i1}(0)|D_i=0}(y) - F_{Y_{i0}(0)|D_i=0}(y).$$

Figure 3 displays the results for both DGPs. In the left panel (DGP 1), the implied counterfactual CDF coincides with the true counterfactual, confirming that the distributional parallel trends condition is satisfied and the DiD approach to the CDF produces a valid probability distribution. In the right panel (DGP 2), the implied and true counterfactual CDFs diverge noticeably. For DGP 2, the implied CDF departs from the true counterfactual $F_{Y_{i1}(0)|D_i=1}(y) = \Phi\big((\ln y - 3.5)/0.5\big)$ because the thin-tailed treated distribution and the heavy-tailed control distribution evolve on different scales. Combining their CDF shifts yields an object that does not match the actual counterfactual. This mismatch is directly linked to the bias in the levels DiD estimator and illustrates how the diagnostic tools of Roth and Sant'Anna (2023) can alert researchers to functional-form sensitivity before estimation.

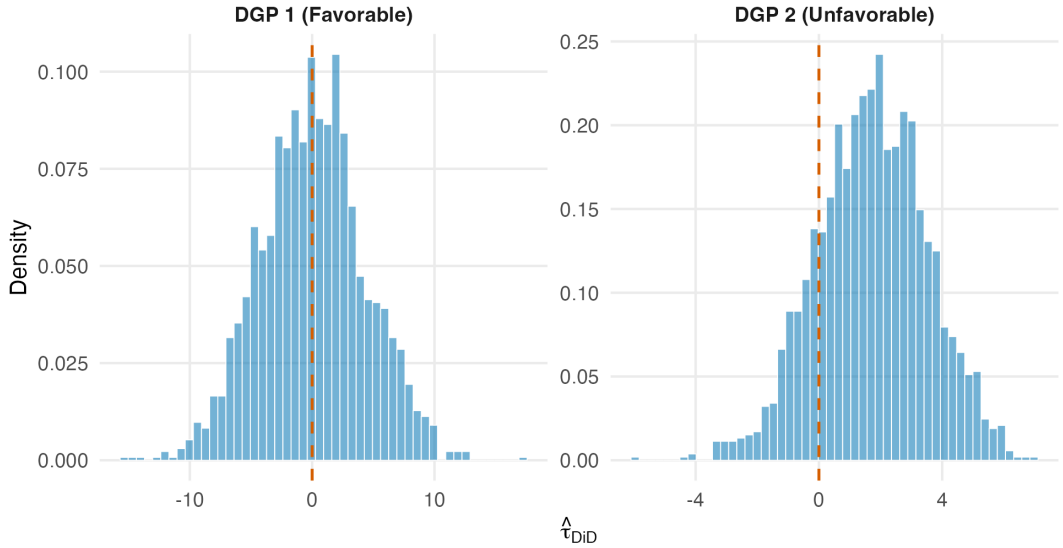
Figure 3. Implied versus true counterfactual CDF for the treated group



Note. Left: DGP 1 (implied and true CDFs coincide). Right: DGP 2 (implied CDF diverges from the true counterfactual, signaling that distributional parallel trends is violated).

Figure 4 summarizes the Monte Carlo evidence by plotting the sampling distributions of the levels DiD estimator across both DGPs. Under DGP 1, the levels estimator is centered at zero, confirming size control. Under DGP 2, the levels estimator is shifted to the right by approximately 1.74, with only modest overlap with the null hypothesis value. This figure provides a compact visual summary of the functional-form-induced bias documented in Tables 1 and 2.

Figure 4. Sampling distributions of the DiD estimator

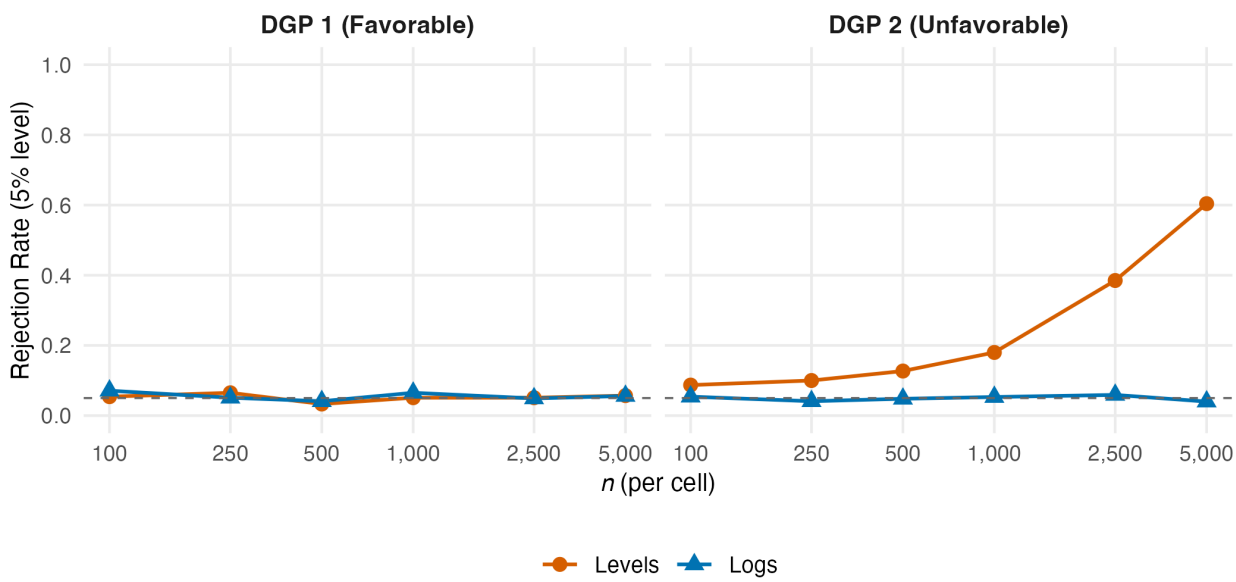


Note. The dashed vertical line marks $\hat{\tau}_{\text{DiD}} = 0$. Under DGP 2, the levels estimator is shifted rightward due to functional-form-induced bias.

Sample-size sensitivity

To assess how the functional form induced bias interacts with sample size, we repeat the DGP 2 simulations for $n \in \{100, 250, 500, 1{,}000, 2{,}500, 5{,}000\}$ per cell with $B = 1{,}000$ replications at each sample size. Figure 5 plots the rejection rate of the 5% Wald test against the sample size on a logarithmic scale. In the logs specification, the rejection rate remains close to 5% across all sample sizes, confirming correct size. In the levels specification, the rejection rate increases monotonically with n: at $n = 100$ the test has limited power to detect the bias, but by $n = 1{,}000$ it exceeds 60%, and at $n = 5{,}000$ it approaches unity. This pattern reflects the fact that the bias is fixed at approximately 1.74 regardless of sample size, while the standard error shrinks at rate $n^{-1/2}$. As a result, functional-form sensitivity that might go unnoticed in small samples becomes increasingly consequential as the sample grows—precisely the regime in which applied researchers are most likely to report statistically significant (but spurious) treatment effects.

Figure 5. Rejection rates of the 5% Wald test as a function of sample size



Note. The dashed horizontal line marks the 5% level. Under DGP 2, the levels rejection rate increases with n because the fixed bias dominates the shrinking standard error.

IV. Critical assessment

Strengths

The paper's primary strength is its theoretical completeness. Rather than offering a partial result or a set of sufficient conditions, Proposition 3.1 provides a full equivalence: PT is robust to all monotonic transformations if and only if the distributional PT condition holds.

Proposition 3.2 adds economic content by translating the abstract distributional requirement into an interpretable mixture structure, making clear what must be true about the data-generating process for invariance to hold. The falsification test is straightforward to implement, and the minimum wage illustration demonstrates its practical value. Together, the theoretical and empirical components form a coherent package that is both rigorous and practically actionable.

Weaknesses

The results rely on a two-period, two-group framework; while the authors note extensions to staggered designs, these are not formally developed in the paper. The mixture representation is an "essentially only if" result, leaving open edge cases where the partition across periods may change, a caveat that could matter in practice but is difficult to verify. Finally, the falsification test only detects violations when the implied counterfactual density turns negative, so it may have limited power against subtle functional-form sensitivity, and as the authors acknowledge, failure to reject does not guarantee robustness.

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